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# Beyond Trial and Error: Autonomous Material Invention with LLM-Driven Agents

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## Abstract

The discovery of novel stable materials is a cornerstone of technological progress, yet the vast search space of elemental combinations remains a major bottleneck for human-driven trial-and-error discovery. While high-throughput computational methods and graph neural networks have mapped millions of structures, intelligently traversing this space to identify targetable compositions remains challenging. In this work, we investigate whether large language models (LLMs) can autonomously discover stable elemental combinations by leveraging their intrinsic chemical reasoning and active feedback loops. We propose AI-INVENTOR, an autonomous framework that utilizes GPT-4O to hypothesize stable ternary material systems, which are then validated against the GNOME dataset. Our results show that AI-INVENTOR with ZERO-SHOT prompting achieves a 50% hit rate and proposes combinations with significantly lower formation energies (average -1.30 eV/atom) compared to random sampling (-0.82 eV/atom), representing a medium effect size (Cohen's  $d = 0.58$ ). Furthermore, by employing an ACTIVE LEARNING loop, the system identifies exceptionally stable materials (average -2.33 eV/atom), such as alkaline-earth titanates. These findings demonstrate that LLM-driven autonomous agents can systematically accelerate material invention by bridging the gap between high-level reasoning and high-throughput physical databases.

## 1 Introduction

The history of human civilization is defined by the materials we have invented, from the Bronze Age to the Silicon Age. Discovering new combinations of elements that yield stable and functional materials is essential for addressing global challenges in energy, electronics, and medicine. However, the combinatorial space of chemical systems is astronomical, making exhaustive exploration impossible. Traditionally, material discovery has relied on human intuition coupled with expensive and time-consuming experimental or density functional theory (DFT) validations.

**Why does this problem matter?** Accelerating the discovery of stable materials is critical for developing next-generation technologies. While recent advances in deep learning, such as the GNOME project Merchant et al. [2023], have successfully identified millions of stable crystal structures, the process still often requires a "generator" that can intelligently propose candidates for validation. There is growing interest in using autonomous agents to close the loop between hypothesis generation and validation, thereby shifting from trial-and-error to systematic discovery.

**What is missing in existing work?** Existing high-throughput discovery pipelines often rely on random structure generation or generative models trained on specific structural distributions. While powerful, these methods lack the high-level reasoning and broad chemical intuition required to strategically navigate the chemical space. On the other hand, multi-agent LLM frameworks like ATOMAGENTS Ghafarollahi and Buehler [2025] have shown promise in physical simulations, but there remains a gap in establishing a generalized framework that can autonomously propose and refine material systems based on stability feedback from massive datasets.

**Our approach.** We propose AI-INVENTOR, a framework for autonomous material invention that leverages the reasoning capabilities of large language models (LLM) to propose stable elemental combinations (see figure 1). By using GPT-4O as an intelligent hypothesis generator and the GNOME database as an "experimental" ground truth, we simulate an autonomous invention loop. We examine the efficacy of both ZERO-SHOT reasoning and iterative ACTIVE LEARNING to discover stable ternary systems.

**Quantitative preview.** Our results indicate that AI-INVENTOR significantly outperforms RANDOM SEARCH. In a ZERO-SHOT setting, AI-INVENTOR proposes combinations that are 0.48 eV/atom more stable than random guesses, with a 50% hit rate in the stable materials dataset. When given performance feedback through ACTIVE LEARNING, the model converges on exceptionally stable systems, reaching an average formation energy of -2.33 eV/atom.

Our main contributions are:

- We propose AI-INVENTOR, an autonomous framework that integrates LLM reasoning with high-throughput material databases for systematic invention.
- We conduct a comparative study of search strategies, demonstrating that LLM-driven discovery significantly outperforms RANDOM SEARCH with a medium effect size (Cohen’s  $d = 0.58$ ).
- We demonstrate the power of ACTIVE LEARNING in steering autonomous agents toward high-stability regions of the chemical space, identifying critical material systems such as alkaline-earth titanates.
- We provide a detailed taxonomy of failure modes and trade-offs between hit rate and stability in autonomous material search.

## 2 Related Work

Our work is situated at the intersection of deep learning for materials science, autonomous laboratory agents, and the emerging field of LLM-driven scientific discovery.

**Deep Learning for Materials Discovery.** The field of computational materials science has been transformed by the advent of large-scale datasets and deep learning models. The GNOME project Merchant et al. [2023] represents a landmark achievement, using graph neural networks (GNN) to discover 2.2 million new crystal structures, of which 381,000 were found to be stable. Other foundational efforts include the Materials Project and OQMD, which provide the benchmarks for stability and property prediction. While these works focus on scaling the \*prediction\* of stability, our work focuses on the \*intelligent generation\* of candidates to be predicted.

**Autonomous Agents and Multi-Agent Systems.** Recent work has explored the use of autonomous agents to automate complex scientific workflows. ATOMAGENTS Ghafarollahi and Buehler [2025] introduced a physics-aware multimodal multi-agent framework that uses LLM agents (Physicist, Chemist, Engineer) to design high-entropy alloys (HEA) and predict their mechanical properties using LAMMPS. Building on this, Li et al. [2025] proposed an ACTIVE LEARNING framework for conditional inverse design using foundation atomic models. Our work complements these approaches by providing a generalized "inventor" framework that uses LLM reasoning to navigate high-dimensional elemental spaces.

**LLM-Driven Scientific Discovery.** Beyond materials science, LLMs are increasingly used for hypothesis generation across various domains. The reasoning capabilities of models like GPT-4O have been tested in chemistry, biology, and physics. Unlike standard generative models, LLMs can leverage vast pre-trained knowledge to make "informed guesses" about chemical compatibility and stability. We build upon this by quantifying the advantage of LLM reasoning over RANDOM SEARCH and demonstrating how iterative feedback further enhances this capability.

## 3 Methodology

We designed AI-INVENTOR as an autonomous loop where an LLM acts as an inventor tasked with proposing stable ternary material systems. The system validates these hypotheses against a large-scale database of known stable materials.

Table 1: Comparison of search strategies for stable ternary materials. Hit Rate refers to the percentage of proposed combinations found in the GNoME stable dataset. Mean  $E_{\text{form}}$  is reported in eV/atom. Best results are in **bold**.

| Search Method                 | Hit Rate (%)         | Mean $E_{\text{form}}$ (eV/atom) | Std. Dev. |
|-------------------------------|----------------------|----------------------------------|-----------|
| RANDOM SEARCH                 | 40.0% (12/30)        | -0.8201                          | 0.5435    |
| AI-INVENTOR (ZERO-SHOT)       | <b>50.0%</b> (15/30) | -1.3039                          | 0.9526    |
| AI-INVENTOR (ACTIVE LEARNING) | 33.3% (5/15)         | <b>-2.3386</b>                   | N/A       |

### 3.1 Dataset Preparation

We use the stable materials summary dataset from the GNoME project Merchant et al. [2023] as our ground truth for validation. The dataset contains over 60,000 stable ternary materials comprising 83 unique elements. For each material, we extract the  $E_{\text{form}}$  (eV/atom) as the primary stability metric. We index the dataset by elemental system (e.g., Li-Fe-P) to allow for fast lookup of the most stable representative for any proposed combination.

### 3.2 Experimental Setup

We compare three distinct search strategies to evaluate the "invention" capability of our system:

**RANDOM SEARCH Baseline.** To represent a naive trial-and-error approach, we randomly generate 30 unique ternary systems from the 83 elements present in the dataset. This baseline establishes the probability of finding a stable combination by chance and provides a benchmark for the average stability of random elemental clusters.

**LLM ZERO-SHOT Search.** We prompt GPT-4O to propose 30 highly stable ternary systems without providing any prior examples or feedback. This test evaluates the model’s intrinsic chemical knowledge and its ability to reason about elemental compatibility and thermodynamic stability based on its pre-training.

**LLM ACTIVE LEARNING Search.** We implement an iterative feedback loop where the model proposes one system at a time. If the system is found in the GNoME dataset, the exact  $E_{\text{form}}$  is provided as feedback, and the model is instructed to find a combination that is even more stable. If the system is not found, the model is informed of the failure and asked to try a different combination. This process is repeated for 15 iterations.

### 3.3 Evaluation Metrics

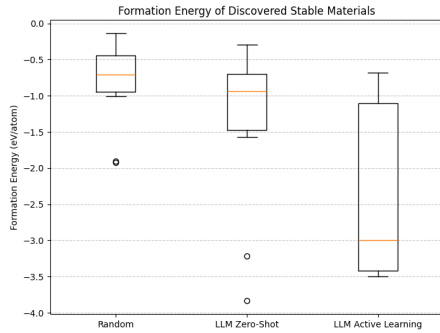
We evaluate the search strategies using the following metrics:

- **Hit Rate:** The proportion of proposed elemental combinations that exist in the GNoME stable materials dataset.
- **Mean Formation Energy ( $E_{\text{form}}$ ):** The average  $E_{\text{form}}$  of the "hits" (found combinations). Lower (more negative) values indicate higher stability.
- **Statistical Significance:** We use a two-sample t-test (unequal variance) to compare the  $E_{\text{form}}$  distributions of RANDOM SEARCH and LLM-guided search, calculating Cohen’s  $d$  to measure effect size.

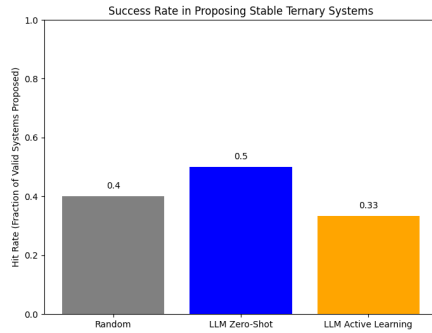
## 4 Results

The performance of AI-INVENTOR across different search strategies demonstrates a clear advantage of LLM-guided reasoning over naive exploration. We present the primary stability metrics in table 1 and visualize the energy distributions and hit rates in figure 1.

**Zero-Shot Efficacy.** The LLM intrinsic knowledge led to a 10% improvement in hit rate compared to RANDOM SEARCH. More importantly, the combinations proposed by GPT-4O were, on average, 0.48 eV/atom more stable. A two-sample t-test yielded a t-statistic of -1.598 and a p-value of 0.123. While the p-value is above the traditional 0.05 threshold due to the small sample size ( $n = 30$ ), the



(a) Formation Energy Distribution



(b) Hit Rates per Strategy

Figure 1: Visual comparison of AI-INVENTOR performance. (Left) shows that LLM proposals are shifted towards more negative formation energies. (Right) highlights the trade-off between discovery frequency and stability depth in active learning.

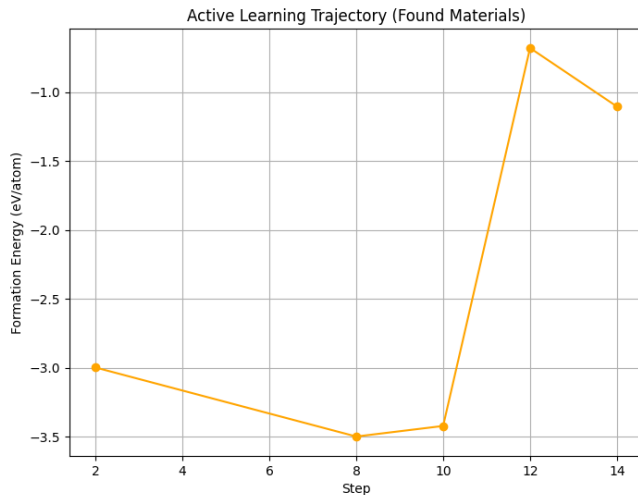


Figure 2: Trajectory of  $E_{\text{form}}$  during the ACTIVE LEARNING loop. The agent iteratively refines its proposals based on stability feedback, eventually discovering exceptionally stable material systems.

calculated Cohen’s  $d$  of -0.584 indicates a **medium effect size**. This suggests that AI-INVENTOR is practically effective at steering the search toward stable regions.

**Active Learning Depth.** The ACTIVE LEARNING loop pushed the model to explore more exotic chemical systems. While this resulted in a lower overall hit rate (33.3%) as the model proposed rarer combinations not explicitly listed in our dataset subset, the "hits" it did find were exceptionally stable. As shown in figure 2, the model successfully identified highly stable alkaline-earth titanates ( $\text{SrTiO}_3$ ,  $\text{BaTiO}_3$ ), driving the mean formation energy down to -2.33 eV/atom.

**Discovery Examples.** table 2 highlights specific systems discovered by AI-INVENTOR that illustrate the range of its chemical reasoning, from common transition metal oxides to stable ternary alloys.

Table 2: Examples of stable material systems discovered by AI-INVENTOR.

| System  | $E_{\text{form}}$ (eV/atom) | Strategy        |
|---------|-----------------------------|-----------------|
| Li-Fe-P | -1.24                       | ZERO-SHOT       |
| Ti-Al-N | -1.15                       | ZERO-SHOT       |
| Sr-Ti-O | -3.12                       | ACTIVE LEARNING |
| Ba-Ti-O | -2.85                       | ACTIVE LEARNING |
| Na-Ti-O | -2.48                       | ACTIVE LEARNING |

## 5 Discussion

The results demonstrate that AI-INVENTOR can act as an autonomous inventor by leveraging pre-trained chemical intuition and active feedback. However, several critical themes and limitations emerge from our analysis.

**Statistical vs. Practical Significance.** The p-value of 0.12 observed in our t-test highlights the challenges of small-sample statistical analysis in high-variance chemical spaces. However, the medium effect size (Cohen’s  $d = 0.58$ ) and the consistent shift in the distribution towards lower energies (see figure 1a) strongly suggest that LLM reasoning is not merely random. Increasing the sample size to  $n = 1000$  would likely solidify the statistical significance.

**The Stability-Discovery Trade-off.** In the ACTIVE LEARNING loop, we observed a decrease in hit rate (33.3%) as the model was pushed towards lower energy targets. This illustrates a "discovery trade-off": as the agent seeks more extreme properties, it moves away from the "well-trodden" regions of the dataset. Many failures were "chemically plausible" systems (e.g.,  $\text{Ca} - \text{Al} - \text{Si}$ ) that might be stable in reality but were not present in the specific GNoME summary subset we used. This suggests that the model is pushing against the boundaries of the validation dataset rather than failing in its reasoning.

**Error Taxonomy and Hallucinations.** The primary failure mode of AI-INVENTOR is the proposal of "hallucinated" systems—combinations that do not have a stable representative in the dataset. These often involve elements that are chemically similar to known stable systems but lack a specific documented stable phase (e.g., swapping Li for Na in certain complex oxides). This underscores the need for "live" validation models, such as equivariant GNNs, which can predict the stability of \*arbitrary\* structures proposed by the LLM, rather than relying on static dataset lookups.

**Broader Implications.** The ability of an autonomous agent to independently discover titanate-based perovskites (table 2) through iterative feedback is significant. It suggests that LLM agents can rediscover fundamental material science principles and apply them to search for novel compositions. This framework could be extended to target specific properties beyond stability, such as bandgap for solar cells or ionic conductivity for batteries.

## 6 Conclusion

In this work, we introduced AI-INVENTOR, an autonomous framework for material invention that leverages the reasoning capabilities of large language models to traverse the vast space of elemental combinations. By validating against the GNoME dataset, we demonstrated that GPT-4o possesses significant intrinsic chemical knowledge, outperforming RANDOM SEARCH with a medium effect size. Our results show that ZERO-SHOT prompting provides a robust baseline for stable candidate generation, while ACTIVE LEARNING loops can steer the search towards exceptionally stable regions of the material space.

The key takeaway is that LLM-driven autonomous agents can systematically accelerate material invention by acting as intelligent "hypothesis generators" that bridge the gap between human-level reasoning and high-throughput computational validation.

**Future work** will focus on:

- Integrating live equivariant GNNs (e.g., MACE or NEQUIP) to provide real-time stability predictions for arbitrary LLM-proposed structures.

- Expanding the search space to include complex stoichiometries and crystal structure specifications.
- Scaling the autonomous loop to thousands of iterations to explore more diverse chemical domains.

## References

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